

# 吴筱琦

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现居城市：伦敦/苏州

意向城市：上海/苏州/杭州 | AI 应用工程师



## 教育背景

帝国理工学院 (QS 2) 计算生命科学研究型硕士

伦敦 2025.09 - 2026.09

- 课程内容：R 语言统计学、生态与进化数据科学、生物计算训练营等统计编程与数据科学核心课程，兼修 GIS、基因组学与生物信息学

帝国理工学院 (QS 2) 生物科学本科

伦敦 2022.09 - 2025.07

- 相关课程：编程与统计，生物信息学，遗传学，细胞生物学与遗传学，应用分子生物学

## 实习经历

上海宣泰医药科技有限公司 分析工程师 · 分析部

上海 2024.07 - 2024.09

- 数据分析：独立完成客户样本的前处理、HPLC 系统操作及数据分析，在严格的质控流程下确保检测结果的准确性和可靠性
- 质量控制：通过严格的质控流程系统识别潜在杂质，为产品质量决策提供关键数据支持，展现了保证数据准确性与可追溯性的工程化思维

## 科研与项目经历

多模态大语言模型提取媒介性状数据评估：高精确率与结构依赖型记录召回

伦敦 2025.12 - 2026.08

Python、Gemma 4 31B IT (多模态 LLM)、PyMuPDF、Docling、提示工程、JSON Schema 校验、二分图记录匹配评估、pytest

- 基准构建：设计加权多样性贪心算法从 VecTraits 超 4 万行、334 篇论文中选取 20 篇论文，构建 2,335 条人工基准记录的开发集与 431 条留出测试集；参照 MIReVTD 标准将 91 字段的原始 schema 精简为 22 字段提取 schema
- 多模态提取流程：基于 PyMuPDF/Docling 构建文本与表格多模态证据包，驱动 Gemma 4 31B IT (256K 上下文多模态大语言模型) 通过提示工程与 JSON Schema 约束完成可溯源性状记录提取；经 11 轮提示词迭代优化，开发集条件宏观字段 F1 达 0.810
- 效果评估与问题定位：设计二分图记录匹配评估框架，按文档级记录遗漏与字段级差异对提取错误系统分类溯源；在留出测试集上评估锁定流程，取得记录精确率 1.000、召回率 0.360、条件宏观字段 F1 0.729，为大模型应用效果的持续优化提供问题定位依据

气候变化对佛罗里达州埃及伊蚊种群动态的影响建模

伦敦 2025.03 - 2025.06

R、deSolve、ggplot2、ODEs、统计验证 (RMSE、Pearson、Spearman)

- 模型适配与验证：将已发表的热带 ODE 房室模型适配至亚热带地理环境，基于 53 周本地监测数据验证模型 (RMSE=1.65, Pearson  $r=0.29$ , Spearman  $\rho=0.30$ )，优化时间滞后与降雨尺度因子

生态建模与种群动力学模拟

伦敦 2024.01 - 2024.02

R、ggplot2、Lotka-Volterra、逻辑斯蒂增长模型

- 建模项目组：构建四个种群动力学模型——集合种群、渔业捕捞 (MSY)、资源竞争及三物种捕食-被捕食模型，均使用差分方程与随机扩展实现；四项均获 90+ 评分 (最高 94 分)，展现扎实的算法实现与参数优化能力

## 技能/其他

- AI/LLM 工程：多模态大语言模型应用开发 (Gemma 4 31B IT)、面向 NLP 信息抽取任务的提示工程与迭代优化、JSON Schema 结构化输出校验、二分图记录匹配评估与错误溯源分析、多后端 LLM 集成 (Gemini API、Ollama 本地推理、OpenAI 兼容接口、HuggingFace 多模态模型)
- Python 与开发工具：Python (主力语言)，Git，虚拟环境，pytest 测试，基于 JSON 配置/状态跟踪的异步子进程编排，Claude Code / Cursor 辅助开发
- Web 开发：FastAPI (路由/服务架构)、React、TypeScript、Vite、Tailwind CSS
- 数据分析与建模：R (ggplot2、dplyr、caper、deSolve)、统计推断 (GLM、PGLS、Wilcoxon 检验、t 检验、Pearson/Spearman 相关)、基于微分方程的动态系统建模
- 语言：英语 (雅思 7.5)，普通话 (母语)，法语 (A2)；兴趣爱好：钢琴，滑雪

# Xiaoqi Wu

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Currently based in: London / Suzhou

AI Application Engineer | Shanghai / Suzhou / Hangzhou

## Education

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**Imperial College London** MSc Computational Methods in Ecology and Evolution    London 2025.09 - 2026.09

- **Curriculum:** Statistics in R, Ecological and Evolutionary Data Science, and Biological Computing Bootcamp as core statistical-programming and data-science modules, alongside GIS, genomics, and bioinformatics

**Imperial College London** BSc Biological Sciences    London 2022.09 - 2025.07

- **Relevant Coursework:** Programming and Statistics, Bioinformatics, Genetics, Cell Biology and Genetics, Applied Molecular Biology

## Internship Experience

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**Shanghai Sunherb Pharmaceutical Technology Co., Ltd.** Analytical Engineer · Analytical Department    Shanghai 2024.07 - 2024.09

- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis on client-supplied samples, ensuring accuracy and reliability under strict quality-control procedures
- **Quality Control:** Systematically identified potential impurities through rigorous quality-control procedures, providing key data to support product quality decisions – reflecting an engineering mindset toward data accuracy and traceability

## Research & Project Experience

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**Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery**    London 2025.12 - 2026.08

*Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, bipartite record-matching evaluation, pytest*

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers (spanning trait types, reporting formats, and data sources) from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set; reduced the 91-column VecTraits schema to a 22-field extraction schema aligned with the MIREVTD reporting standard
- **Multimodal Extraction Pipeline:** Built a multimodal evidence pipeline combining PyMuPDF full-text extraction, Docling table-structure parsing, and 150 DPI figure/table rendering, fed into Gemma 4 31B IT (a 256K-context multimodal LLM) to extract provenance-linked trait records from PDF text, tables, and images via prompt engineering and JSON Schema-constrained output; iterated across 11 prompt versions, with the final version reaching a development-set conditional macro field F1 of 0.810
- **Effect Evaluation & Issue Diagnosis:** Designed a bipartite record-matching evaluation framework (row-set, conditional-field, and end-to-end field metrics) and systematically classified extraction errors into document-level record omissions versus field-level disagreement; evaluated the locked pipeline on the held-out test set, achieving row precision of 1.000, row recall of 0.360, and conditional macro field F1 of 0.729 – confirming a "high-precision, structure-dependent recall" profile and providing actionable diagnostics for continued optimization of the model's application performance

**Modelling the Impact of Climate Change on *Aedes aegypti* Population Dynamics in Florida**    London 2025.03 - 2025.06

*R, deSolve, ggplot2, ODEs, statistical validation (RMSE, Pearson, Spearman)*

- **Model Adaptation & Validation:** Adapted a published tropical ODE compartment model to a sub-tropical geography and validated it against 53 weeks of local mosquito-surveillance data (RMSE=1.65, Pearson  $r=0.29$ , Spearman  $\rho=0.30$ ), optimizing time-lag and a rainfall scaling factor

**Ecological Modelling and Population Dynamics Simulation**    London 2024.01 - 2024.02

*R, ggplot2, Lotka-Volterra, logistic growth model*

- **Modelling Suite:** Built four population-dynamics models – metapopulation, fisheries harvesting (MSY), resource competition, and a 3-species predator-prey system, each implemented with difference equations and stochastic extensions; all four scored 90+ (up to 94), demonstrating solid algorithm implementation and parameter-optimization skills

## **Skills & Other**

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- **AI/LLM Engineering:** Multimodal LLM application development (Gemma 4 31B IT), prompt engineering and iterative optimization for NLP information-extraction tasks, JSON Schema-constrained structured output, bipartite record-matching evaluation and error diagnostics, multi-backend LLM integration (Gemini API, Ollama local inference, OpenAI-compatible APIs, HuggingFace multimodal models)
- **Python & Tooling:** Python (primary language), Git, virtual environments, pytest testing, async sub-process orchestration with JSON-based config/state tracking, Claude Code / Cursor for AI-assisted development
- **Web Development:** FastAPI (routers/services architecture), React, TypeScript, Vite, Tailwind CSS
- **Data Analysis & Modelling:** R (ggplot2, dplyr, caper, deSolve), statistical inference (GLM, PGLS, Wilcoxon signed-rank, t-tests, Pearson/Spearman correlation), ODE-based dynamic system modelling
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, skiing